

W6 Exercises



Consult W5 TA session notes if you run into any problems! Don't turn to **brute force**!

Task 0

- Download and import the dataset `ames`

Task 1: Housing Prices in 2026

- Set seed as `2026`
- Use `paste()` and `for` loop to create a character vector `date_2026` that contains 365 days in 2026, with each date repeats 3 times. i.e., `c("2026-01-01", "2026-01-01", "2026-01-01", ..., "2026-12-31")`
 - **Hint:** use `if (month %in% vec)` to check if there are 28, 30, or 31 days in that month (`%in%` checks if the former is a member of the latter)
- Use `as.Date()` to convert `date_2026` as `Date` data
- Create a numeric vector `housing_price` with length 1095 ($=365*3$) and have the value generated from a normal distribution with `mean = 10000` and `sd = 2000`
- Combine `date_2026` and `housing_price` as a new dataframe `housing_2026`
- Use `geom_line()` with `linewidth = 0.3` to plot the average housing price for each date in 2026